Three requirements on a relativistic home for MOND phenomenology

Carl P. Zimmerman[1]

[1] *Briar Creek Tech*

(Dated: 18 August 2026 (v3))

Aether–scalar–tensor gravity (AeST) [Skordis & Złośnik, Phys. Rev. Lett. **127**, 161302 (2021)] is the only relativistic MOND-class theory known to reproduce the cosmic microwave background power spectrum, and it contains a single free function $\mathcal{F}(\mathcal{Y},\mathcal{Q})$ that its authors leave unspecified. We show that specifying it is obstructed, and we extract from the obstruction three necessary conditions on *any* relativistic completion of MOND phenomenology. Each is established by an explicit construction that satisfies the previous conditions and fails the next. **(R1)** The free function must depend on the gradient of the *total* potential, not on the scalar's own. With $\mathcal{F}(\mathcal{Y})$ the quasi-static reduction forces the anomalous acceleration $U(y)$ to be monotone increasing, hence to saturate at some $U \to s$, hence to produce a *constant* sunward anomaly $s\,a_0$ at every planetary distance. Perihelion precession then requires $s \leq 1.27 \times 10^{-5}$ while the radial acceleration relation requires $s \geq 0.435$ — incompatible by 1.2–3.4 $\times 10^4$. There is no external-field relief: computed for this family rather than imported, it is 1.000000×. **(R2)** Eating the total gradient must not destabilise the aether's vector sector. Replacing $\mathcal{Y}$ by $Z \equiv J^\mu J_\mu$, the squared aether acceleration, satisfies R1 exactly — the quasi-static reduction becomes AQUAL, the exponential interpolation becomes legal, $\gamma_{\rm PPN} = 1$ and G_N stay unchanged, the 1 AU anomaly falls to $10^{-3458.7}\,\mathrm{m\,s^{-2}}$, the FRW background is *exactly* undisturbed, and the derived $a_0(z)$ law survives bit-for-bit. It fails on the vector modes, and it fails worst exactly where it was invoked: the vector kinetic coefficient is $C_V = K_B - (2-K_B)J_Z$, and the same screening that voids the gap drives $J_Z \to 1$, giving $C_V = -1.50$ and $c_V^2 = -0.167$ at $K_B = 0.25$ — a ghost with a gradient instability whose growth rate $0.408\,ck$ is unbounded in the ultraviolet. The unstable region covers the entire solar system and Oort cloud ($r < 5.16 \times 10^4$ AU) and the entire optical body of a galaxy ($r < 79$ kpc), e-folding in 624 yr even at the longest wavelength for which the reduction is controlled. **(R3)** And the repair may not be bought with a split between the cosmological and local gravitational constants. Rescaling, $\mathcal{J} \to s\mathcal{J}$, cures the ghost at every acceleration for $s < K_B/(2-K_B)$ — and forces $\tilde{G}/G_N \leq 0.121$, giving a primordial helium abundance $Y_p = 0.086$ against an observed 0.2449 ± 0.0040 (-42σ) and a gravitational radiation density *below that of the photons alone*. Gravitational waves do not pay this price: the standard-siren amplitude's sensitivity is exactly zero. We also record a structural identity: AeST's action already contains an explicit $+(2-K_B)a^\mu a_\mu$, so its apparent $c_4 = 0$ is an artifact of variables; and $c_T = 1$ holds exactly, for any free function.

I. SETUP

AeST's action, verified against the authors' source [1, 2], is

$$S = \int d^4x\, \frac{\sqrt{-g}}{16\pi\tilde{G}} \Big[R - 2\Lambda - \tfrac{K_B}{2} F^{\mu\nu}F_{\mu\nu} + 2(2-K_B)J^\mu\nabla_\mu\varphi - (2-K_B)\mathcal{Y} - \mathcal{F}(\mathcal{Y},\mathcal{Q}) - \lambda(A^\mu A_\mu + 1)\Big] + S_{\rm m}[g], \tag{1}$$

with $\mathcal{Q} = A^\mu\nabla_\mu\varphi$, $\mathcal{Y} = (g^{\mu\nu} + A^\mu A^\nu)\nabla_\mu\varphi\nabla_\nu\varphi$, $F_{\mu\nu} = 2\nabla_{[\mu}A_{\nu]}$, and $J^\mu = A^\nu\nabla_\nu A^\mu$ the aether's acceleration. Matter couples to $g_{\mu\nu}$ alone.

Throughout we use the acceleration scale

$$a_0 = \kappa c\sqrt{G\rho_\Lambda} = 9.3619 \times 10^{-11}\,\mathrm{m\,s^{-2}}, \tag{2}$$

with an alternative footing giving 1.1279×10^{-10}; both are carried throughout. The coefficient $\kappa = \frac{1}{2}$ is *fitted, not derived*: three determinations give 0.551 ± 0.043, 0.465 ± 0.076 and 0.537 ± 0.071, combining to 0.529 ± 0.034. Write $y = g_{\rm bar}/a_0$ and $U(y) = u/a_0$ for the anomalous acceleration.

II. R1: THE FREE FUNCTION MUST EAT THE TOTAL GRADIENT

With $\mathcal{F}(\mathcal{Y})$, the quasi-static gradient sector diagonalises: $\Psi = \Psi_N + \chi$ with $\nabla^2\Psi_N = 4\pi\hat{G}\rho$, and in spherical symmetry Gauss's theorem gives the local algebraic law $u\,J_Y(u^2) = g_{\rm bar}$ with $u = |\nabla\chi|$. The scalar's own gradient *is* the anomalous acceleration. Reconstructing the free function from a chosen interpolation gives $J_Y(Y) = y(U)/U$, invertible iff U is strictly increasing; equivalently the longitudinal eigenvalue of $M_{ij} = J_Y\delta_{ij} + 2J_{YY}u_iu_j$ is positive under the same condition. A non-monotone U means both a multi-valued $\mathcal{F}$ and a longitudinal gradient ghost.

This is specific to AeST. In AQUAL [3] the free function depends on the total potential's gradient and stability requires only $dg_{\rm obs}/dg_{\rm bar} > 0$ — satisfied by the exponential interpolation $\nu = 1/(1 - e^{-\sqrt{y}})$ [4], minimum 0.968. The same kernel is legal in one theory and fatal in the other, purely from which gradient the free function eats.

Monotonicity plus $U/y \to 0$ forces saturation, $U \to s$: a constant sunward anomaly $s\,a_0$ at every planetary distance. By the Gauss planetary equations a constant radial perturbation gives $\dot{\varpi} = s\,a_0\sqrt{1-e^2}/(na)$; against anomalous-precession limits [5, 6] the worst planet requires

$$s \le 1.27\times 10^{-5}\ \text{(canonical)},\quad 1.05\times 10^{-5}\ \text{(alt)}. \tag{3}$$

Applying $U(2) \ge 0.4$ to the family $J_Y = v/(1-v/s)$, $v = \sqrt{Y}/a_0$, gives $s \ge 0.4348$; relaxing to a global fit (rms ≤ 0.15 dex on 3389 SPARC points [7], Υ in the Spitzer band) gives 0.157–0.294 depending on treatment. Hence

$$\boxed{\text{GAP: } 1.2\text{–}3.4\times 10^4.} \tag{4}$$

Every candidate relief was computed and none helps. *The external-field effect contributes nothing*: derived for this family by two independent routes (an $\ell = 1$ penetration ODE and a flux bound requiring no perturbation theory), the relief on the saturated anomaly is 1.000000× — the external field screens *itself*, not the anomaly, and somewhere on the 1 AU sphere the anomaly is at least $s\,a_0(1-4\times10^{-9})$. A density-dependent a_0 *hurts*: with suppression f the constrained product is sf, and $f\,U_s(2/f) = 0.4$ gives 0.435 at $f = 1$ but 2.00 at $f = 0.1$, unsatisfiable below $f = 0.080$ because $U \le \sqrt{y}$ caps the family. Solving the non-spherical problem for a Miyamoto–Nagai disc raises the required floor by 1.07–1.10. Refitting Υ buys 1.34×.

III. R2: EATING THE TOTAL GRADIENT MUST NOT DESTABILISE THE VECTOR SECTOR

R1 says: use the total gradient. AeST already contains the object that supplies it. With $D^\mu = q^{\mu\nu}\nabla_\nu\varphi$ one has the identity, verified three times independently and requiring no computer algebra,

$$\begin{aligned}&2(2-K_B)J\cdot\nabla\varphi - (2-K_B)\mathcal{Y}\\ &\qquad\equiv +(2-K_B)Z - (2-K_B)|D-J|^2,\end{aligned} \tag{5}$$

with $Z \equiv J^\mu J_\mu$. *AeST therefore already contains an explicit* $+(2-K_B)a^\mu a_\mu$*: its apparent* $c_4 = 0$ *is an artifact of variables.* Moreover $J^i = \nabla^i\Psi$ with coefficient exactly unity, with Z's leading coefficient independent of φ, Φ and the aether perturbation — so Z *is* the total potential gradient squared.

Replacing $\mathcal{F}(\mathcal{Y},\mathcal{Q})$ by $\mathcal{F}(Z,\mathcal{Q})$ then works exactly as R1 requires. The aether-longitudinal equation becomes $v = \nabla\Psi$ pointwise, the $(1+J_Y)$ factor disappears, and the reduction is *AQUAL exactly*, with $\mu(g_{\rm obs}) = J_Z(g^2_{\rm obs})$ and $\hat{G}$ unchanged. The exponential kernel becomes legal (min $dx/dy = 0.968$, against min $dU/dy = -0.032$ under $\mathcal{F}(\mathcal{Y})$ for the same kernel), and the 1 AU anomaly becomes $10^{-3458.7}\,\mathrm{m\,s^{-2}}$ — some 10^{3445} below the Sereno–Jetzer bound [8], replacing an $s\,a_0$ that sat 1279× above it. Further, $c_T^2 = 1$ exactly for any free function, and G_N is finite and equal to the unmodified value.

This construction does everything R1 asks, and several things it does not. The FRW background is *exactly* undisturbed: the comoving aether congruence is geodesic, so $J^\mu = 0$ identically for arbitrary lapse, hence $\bar{Z} = 0$ and $\mathcal{F}(0,\mathcal{Q}) = K(\mathcal{Q})$ — the same background function AeST's own $\mathcal{F}(\mathcal{Y},\mathcal{Q})$ reduces to — and the new term's stress contribution carries at least one factor of J and vanishes too. Friedmann, $w = -1$, the dust mode, $Q(a)$ and therefore the derived $a_0(z)$ law are bit-for-bit the AeST objects. In the static sector one finds AQUAL with $\mu = J_Z$, $\gamma_{\rm PPN} = 1$ and $G_N = \hat{G}$ finite. At linear order $J_i = \nabla_i E$ exactly, with $E = \dot\alpha + \Psi$ the theory's own mixing variable, so $Z = |\nabla E|^2/a^2$ is second order; and since $F^{\mu\nu}F_{\mu\nu} = -2Z$ at $O(\delta^2)$, any analytic linear-in-Z piece of $\mathcal{F}$ is a mere relabelling $K_B \to K_B - \mathcal{F}_Z(0)$, with $\mathcal{F}_Z(0) = 0$ on the determined family because every interpolation's deep-MOND end has $\mu_{\rm phys} \to 0$.

Its cosmological cost is real but payable, and we state it because it is the one place the framework's own machinery does unexpected work. Eating the *total* gradient switches the MOND branch on at recombination, where a CLASS run puts the acoustic-band gravitational acceleration at only $y = 7.6$–11.9 absent running; the resulting $\epsilon = 1-\mu(y) \simeq 1/2y$ is a few per cent kernel-robustly, worth 18–112σ aggregated over $\ell = 2$–2500 at unit response. The CMB tolerates it only if $a_0(\mathrm{rec})/a_0(0) \le 9.7\times10^{-3}$ to 3.7×10^{-1} across eight forks — and the suppression this framework independently derives, 0.0060, clears the tightest fork and the loosest, neither number having been tuned to the other. (Delivering it needs $\nu_0 \ge 5.6\times10^{-9}$ to 8.2×10^{-6}, which exceeds the independent RAR ceiling 2.36×10^{-6} on two of eight forks — exactly the $\alpha = 1$ kernel under the conservative reading — while the exponential kernel this construction relegalises clears on every reading. The interlock scales as S^2 in the uncomputed CMB response $S = |d\ln C_\ell/d\epsilon|$.)

It dies on the vector modes, and it dies worst exactly where it was invoked. The transverse-vector kinetic coefficient is

$$C_V = K_B - (2-K_B)J_Z,\qquad c_V^2 = K_B/C_V, \tag{6}$$

so the theory is ghost-free only for $J_Z < K_B/(2-K_B)$. But $J_Z = \mu_{\rm phys}$ is precisely the interpolation function, and the Newtonian screening that deletes the ephemeris gap is the statement $\mu_{\rm phys} \to 1$. At $K_B = 0.25$ this gives $C_V = -1.50$ and $c_V^2 = -0.167$: a ghost carrying a gradient instability, growth rate $0.408\,ck$, unbounded in the ultraviolet. Converting $J_Z < K_B/(2-K_B)$ into a radius, the unstable region covers the entire solar system and Oort cloud ($r < 5.16\times10^4$ AU canonical, 4.70×10^4 alt) and the entire optical body of a $10^{11}M_\odot$ galaxy ($r < 79.2$ kpc canonical, 72.1 alt). Even at the longest wavelength for which the reduction is controlled, $\lambda = c^2/g$ at 1 AU, the e-folding time is 624 yr. No cutoff choice saves it, and only $K_B \ge 1$ removes the bound at all.

The structure of the failure is the point: *the same factor* $\mu_{\rm phys} \to 1$ *that makes the solar-system screen-*

ing work is what makes C_V most negative. Escaping R1 and destabilising the vector sector are one operation, not two. AeST itself, with $\mathcal{F}(\mathcal{Y})$ and no Z-term, has $C_V = K_B > 0$ and is healthy here; the liability belongs to this construction alone.

IV. R3: AND THE REPAIR MAY NOT BE BOUGHT WITH A $\tilde{G}/G_N$ SPLIT

Equation (6) names its own repair. Both sector conditions constrain only the total Z-coefficient: static attraction needs it positive, vector no-ghost needs it below K_B. Rescaling the free function, $J \to sJ$, satisfies both at every acceleration for $s < K_B/(2-K_B)$; the window is $0 < s < 0.138$ at $K_B = 0.25$, and the solar-system screening is untouched, being s-independent.

The price is that with $\mathcal{F}(Z)$ the non-$\mathcal{F}$ static Lagrangian is annihilated identically by $v = \nabla\Psi$, making the free function's normalisation the sole source of Newton's constant: $G_N = \hat{G}/s$. The no-ghost condition is therefore equivalent to $2\tilde{G} < K_B G_N$, i.e.

$$\tilde{G}/G_N \leq 0.121 \ (K_B = 0.25), \qquad 0.048 \ (K_B = 0.10), \tag{7}$$

against AeST's 0.875 — an 8.3× to 20.7× split. It cannot be paid.

Big-bang nucleosynthesis. $Y_p = 0.086$ at the no-ghost ceiling against an observed 0.2449 ± 0.0040: -42σ from the same calculation's $\tilde{G}/G_N = 1$ control. A closure-free bound assuming no nucleosynthesis model at all still gives $Y_p \leq 0.176$, i.e. -17σ.

The CMB, and this is an inconsistency rather than a tension. The gravitational sector's radiation density comes out at 0.2046 of the thermal photon density — *below the photons alone*, whose value FIRAS [9] fixes with no gravity anywhere in the chain. The equivalent $N_{\text{eff}} = -3.503$. The hard floor is missed by 4.9× ($K_B = 0.25$) to 12.2× ($K_B = 0.10$), and the ceiling sits 139σ from viability in cosmic-variance units.

Gravitational waves do not pay this price, and we state that against our own interest. The tensor sector is exactly Einstein–Hilbert: on FRW plus a transverse-traceless (TT) mode, $J^\mu = 0$, $Z = 0$, $F_{\mu\nu} = 0$ and $\mathcal{Y} = 0$ all hold identically and to all orders in h, so the dark sector carries no anisotropic stress for any free function, giving $c_T = 1$ and $\alpha_M = 0$ exactly. The standard-siren amplitude's sensitivity to s is then exactly zero: $\tilde{G}$ enters the strain and G_N the orbit, and they cancel in d_L^{GW}. GW170817 [10] places *no bound* on s. Two liabilities the same derivation creates are recorded but *not scored*: the inferred chirp mass rescales as $r^{3/5}$ ($4.13\,M_\odot$ for GW170817 at the ghost edge, which needs the theory's own strong-field stellar structure to close), and the tensor-channel orbital decay is $\dot{P}_b/\dot{P}_b^{\text{GR}} = r$, an 8× deficit against B1913+16 [11]. We decline to score the second because the identical ledger applied to *unmodified* AeST would exclude it at 77σ — the tensor-only flux ledger is evidently incomplete, and the total radiated flux is not computed here. The usable result is a constraint on the shape of any repair: the missing channels must supply the same $1-r$ fraction at every orbital period, since $\dot{P}_b/\dot{P}_b^{\text{GR}}$ is measured to 0.16% in a 7.75 h orbit and to 1.3×10^{-4} in a 2.45 h one [12], and a dipole enters at relative order $(c/v)^2$ and cannot be period-independent.

V. TESTABLE PREDICTIONS

A no-go is tested by construction rather than by observation, but the derivations above carry four consequences that data can decide.

P1: $\alpha_M = 0$ and $c_T = 1$, exactly, for any free function. This is the strongest prediction here and the one least dependent on anything we have chosen. On FRW plus a TT mode, $J^\mu = 0$, $Z = 0$, $F_{\mu\nu} = 0$ and $\mathcal{Y} = 0$ hold identically and *to all orders in h*, so every non-Einstein invariant is h-independent and the dark sector's TT stress is $T^x{}_x - T^y{}_y = 0$. The tensor sector is therefore exactly Einstein–Hilbert with $M_*^2 = 1/8\pi\tilde{G}$ constant. AeST and every construction in this paper are consequently indistinguishable from general relativity in gravitational-wave propagation — no running Planck mass, no modified friction, no c_T deviation, at any redshift and for any K_B, s or a_0 footing. GW170817's $|c_T - 1| \lesssim 10^{-15}$ is passed trivially rather than fitted. *A confirmed $\alpha_M \neq 0$ from LISA or Einstein Telescope standard sirens falsifies this entire class in one measurement*, and cannot be absorbed by any choice of free function.

P2: the orbital-decay deficit must be period-independent. If the tensor-only flux ledger of Sec. IV is incomplete, as we argue, the missing channels must supply the same $1-r$ fraction of the flux at every orbital period. A dipole channel enters at relative order $(c/v)^2$ and therefore cannot be period-independent; the repair must renormalise the quadrupole coefficient itself. Any future binary showing a period-*dependent* deviation from the GR quadrupole formula excludes the repair, and with it the aether-vector rescaling.

P3: the $\mathcal{F}(\mathcal{Y})$ class predicts a constant sunward anomaly with a fixed planet ordering, $\dot{\varpi} = s\, a_0\sqrt{1-e^2}/(na)$, falling with semi-major axis in a way no fitted $\dot{\varpi}$ per planet can mimic, since a single s sets all of them. This is already excluded (Sec. II), which is *why* R1 exists; a future ephemeris solution reporting a coherent residual of this shape would revive rather than kill it.

P4: the Z-form predicts no solar-system deviation at all, at any measurable level: the 1 AU anomaly is $10^{-3458.7}\,\text{m}\,\text{s}^{-2}$ and is s-independent. Any claimed solar-system MOND detection excludes it outright.

These are separate from framework-level predictions that this paper leaves untouched, since they belong to the normalisation (2) rather than to any completion: a hash-frozen wide-binary velocity ratio decided by Gaia DR4; a declining $a_0(z)$, which puts the MOND regime *off* at recombination at 0.0060 of its present value and is falsified by any robust a_0 evolution below $z \sim 5$; and $\nu_0 \leq 2.36 \times 10^{-6}$, i.e. no environmental variation of a_0 at a level current data could see.

VI. WHAT IS AND IS NOT CLAIMED

Claimed. The three requirements, each with an explicit construction that satisfies its predecessors and fails it. The identity (5) and its corollary that AeST's $c_4 = 0$ is a choice of variables — and, more sharply, that $c_4^{\rm eff}$ is *sector-dependent*, so "$c_{14} = 2$" is not a well-defined statement about this theory. That $c_T = 1$ holds exactly in AeST and in the $\mathcal{F}(Z)$ theory for any free function. That the monotonicity obstruction of R1 is genuinely escapable, and what escaping it costs. As a subsidiary but independent result: that the Z-form's cosmological cost is met by the framework's own derived $a_0(z)$ suppression on all eight forks — which does not save it, R2 being fatal, but does establish that the CMB was never the binding constraint on this construction.

Not claimed. That the requirements are sufficient — they are necessary conditions extracted from three failures, and a completion satisfying all three is not thereby viable. That the Z-form's CMB verdict is settled: it is conditional on an uncomputed response and on two of eight forks, and is in any case not what kills it. That the scalar dynamical modes, α_1 or α_2 have been computed for the Z-form; they have not. That $\kappa = \frac{1}{2}$ is derived; it is fitted, and must be quoted with its H_0, the distance-free determination carrying an unpriced $\sim 7\%$ systematic. That dark matter is absent: $\Omega_{\rm dm}$ is full here and the claim is "no dark-matter *particle*". Whether the dark sector's dust remains bound inside galaxies is unresolved and untouched by any of this — it is a $\mathcal{Q}$-sector problem, and all three constructions modify the $\mathcal{Y}$-sector.

Unaffected by any of the above, being independent of which completion carries them: the normalisation (2), the radial acceleration relation at 0.108 dex on 175 SPARC galaxies, weak lensing from 40 kpc to 2.2 Mpc with no dark component, the baryonic Tully–Fisher relation, and a hash-frozen wide-binary prediction decided by Gaia DR4.

What a viable completion must supply. A screening mechanism that is not a non-monotone interpolation (R1); one whose approach to the Newtonian limit does not simultaneously drive a kinetic coefficient negative (R2); and one that does not separate the cosmological and local gravitational constants (R3). R2 is the sharpest of the three as a design constraint, because it is not a statement about any particular invariant: it says that whatever object the free function eats, the *same limit* that produces Newtonian screening must not be the limit that destabilises another sector. Superfluid constructions [13], in which the MOND force exists only inside a condensate phase and the Newtonian exterior is a different phase rather than a limit of the same function, evade that coupling by construction and are the obvious next candidate.

A note on the process. Version 1 of this paper attributed the Z-form's death to a cosmological degeneracy $\mathcal{F}_Z(0) = K_B$; version 2, correcting that, over-swung and reported the construction as viable. Both were wrong. The first misread "degenerate with the K_B term" (indistinguishable from it) as "the equation degenerates"; the second, having removed a false death, failed to check that a true one was already on record in the vector sector. The correct statement is the one above: the cosmology is payable, and the vector ghost is fatal. Both errors are recorded, dated, with their directions stated, in `RETRACTIONS.md`.

ACKNOWLEDGMENTS

Every number in this paper is produced by a committed self-checking script at https://github.com/carlzimmerman/zimmerman-formula; each prints numbered checks and exits non-zero on failure. Withdrawn claims are recorded, dated, in `RETRACTIONS.md` — including one in this paper's own drafting, where a gravitational-wave exclusion was written from an unverified number and corrected against the committed script before filing. The contribution here is the three requirements, the identity (5), and the constructions that establish them — not the underlying theory, which is Skordis and Złośnik's.

[1] C. Skordis and T. Złośnik, Phys. Rev. Lett. **127**, 161302 (2021), arXiv:2007.00082.
[2] C. Skordis and T. Złośnik, arXiv:2109.13287.
[3] J. Bekenstein and M. Milgrom, Astrophys. J. **286**, 7 (1984).
[4] M. Milgrom and R. H. Sanders, Astrophys. J. **678**, 131 (2008), Eq. (13) at $\alpha = \frac{1}{2}$.
[5] E. V. Pitjeva and N. P. Pitjev, Mon. Not. R. Astron. Soc. **432**, 3431 (2013).
[6] A. Fienga *et al.*, Celest. Mech. Dyn. Astron. **111**, 363 (2011).
[7] F. Lelli, S. S. McGaugh, and J. M. Schombert, Astron. J. **152**, 157 (2016).
[8] M. Sereno and Ph. Jetzer, Mon. Not. R. Astron. Soc. **371**, 626 (2006).
[9] D. J. Fixsen *et al.*, Astrophys. J. **473**, 576 (1996).
[10] B. P. Abbott *et al.* (LIGO/Virgo), Phys. Rev. Lett. **119**, 161101 (2017).
[11] J. M. Weisberg and Y. Huang, Astrophys. J. **829**, 55 (2016).
[12] M. Kramer *et al.*, Phys. Rev. X **11**, 041050 (2021).
[13] L. Berezhiani and J. Khoury, Phys. Rev. D **92**, 103510

(2015).
[14] Reproduced by `real_research/reviews/opt1_cmb_2026.py` (36/36 checks), which carries the CLASS run, the eight-fork table and the S-threshold.
[15] B. Z. Foster and T. Jacobson, Phys. Rev. D **73**, 064015 (2006).